

Carryover Predictor Comparison in pmsimstats2025: A `tsd` Computation Bug and the Original's Open Power Question

pmsimstats2025 Project Notes

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1 Context

The `carryover_focus/` simulation isolates the effect of pharmacological carryover on statistical power for biomarker moderation detection in the Hybrid N-of-1 aggregated trial design [1]. The simulation uses Hendrickson's binary drug exposure predictor `Dbc` (bounded $[0, 1]$), which equals 1 when on drug and decays exponentially when off drug.

This document reports two findings:

1. The `pmsimstats2025` reimplementation computes the time-since-discontinuation variable (`tsd`) differently from Hendrickson's original `buildtrialdesign()` function. The 2025 code produces $t_{sd} = 0$ at every first off-drug measurement, while the original produces $t_{sd} \geq 1$ (the interval width leading into that measurement). This discrepancy causes a step-function power cliff in the 2025 simulation that does not exist in the original framework.
2. Independent of this bug, the original Hendrickson simulation shows power declining more rapidly with increasing carryover half-life than pharmacological reasoning would predict. The cause of this behavior remains open and requires investigation after the `tsd` computation is corrected to match the original.

2 The Two Analysis Models

The simulation compares two model specifications, toggled by the `model_carryover` flag. Both follow Hendrickson et al.'s [1] binary predictor approach:

Model A (`model_carryover = TRUE`):

```
response ~ Dbc * bm_centered + week
```

Model B (`model_carryover = FALSE`):

```
response ~ treatment * bm_centered + week
```

Both are fitted with `nlme::lme()` using a random intercept per participant and a continuous-time AR(1) correlation structure (`corCAR1(form = ~ week | participant_id)`), which models the decay of within-subject correlation as a function of the time gap between measurements. The difference between the two models lies entirely in the drug exposure predictor used in the interaction term.

2.1 Model Variable Definitions

response

The simulated clinical outcome for each participant at each measurement week. In the DGP, it is the sum of baseline status, biological response to drug (BR), expectancy response (ER), and time-in-trial trend (TR): $\text{response} = \text{baseline} + \text{BR} + \text{ER} + \text{TR}$.

week

The calendar week of the measurement within the trial schedule (e.g., 4, 8, 9, 10, 11, 12, 16, 20 for the Hybrid design). Included in the model as a fixed-effect covariate to absorb the secular time trend.

bm_centered

The participant’s biomarker value, standardized to zero mean and unit variance across the sample: $\text{bm_centered}_i = (\text{biomarker}_i - \bar{\text{biomarker}}) / \text{SD}(\text{biomarker})$. The drug-by-biomarker interaction term tests whether participants with higher biomarker values respond differently to treatment.

participant_id

Unique identifier for each participant ($i = 1, \dots, 70$). Defines the grouping structure for the random intercept and the within-subject correlation.

treatment

Binary indicator: 1 if the participant is on drug at that measurement week, 0 if off drug. Determined by the Hybrid design’s path assignment (4 paths with staggered discontinuation and reinitiation schedules). In Model B, this serves as the “naive” drug exposure predictor: it assumes zero carryover.

2.2 The Dbc Predictor

The drug exposure predictor Dbc (“drug binary with carryover”) follows Hendrickson et al. [1]. It is bounded in $[0, 1]$:

$$\text{Dbc} = \begin{cases} 1 & \text{if on drug} \\ 0 & \text{if off drug and } t_{1/2} = 0 \quad (\text{no carryover}) \\ \left(\frac{1}{2}\right)^{t_{\text{sd}}/t_{1/2}} & \text{if off drug and } t_{1/2} > 0 \end{cases}$$

Here t_{sd} is the time since discontinuation in weeks and $t_{1/2}$ is the carryover half-life (the time for the residual drug effect to decay to half its value at discontinuation). The critical question is how t_{sd} is computed; the original and the 2025 reimplementation differ on this point (Section 4).

3 Hendrickson’s Original tsd Computation

3.1 The buildtrialdesign() Algorithm

In the original pmsimstats package, `tsd` is computed by `buildtrialdesign()` (`buildtrialdesign.R`, lines 86–106). The algorithm accumulates off-drug interval widths:

```

t_wk <- c(4, 4, 1, 1, 1, 1, 4, 4) # interval widths
tsd <- t_wk * (od != 1)           # seed: width if off-drug
for (iTP in 2:nP) {
  if (od_intervals[iTP] > 0)      # on drug
    tod[iTP] <- tod[iTP] + tod[iTP-1]
  else                            # off drug
    tsd[iTP] <- tsd[iTP] + tsd[iTP-1]
}
tsd <- everondrug * tsd

```

The implicit assumption is that drug discontinuation occurs at the *start* of the interval, so by the measurement at the end of the interval the participant has been off drug for the full interval duration. The first off-drug measurement therefore has $t_{sd} = t_{wk}$ (the interval width), never zero.

3.2 Original tsd Values for All Four Paths

The Hybrid design uses measurement weeks (4, 8, 9, 10, 11, 12, 16, 20) with interval widths $t_{wk} = (4, 4, 1, 1, 1, 1, 4, 4)$. The four **ondrug** vectors from the publication vignette:

```

Path A: c(1, 1, 1, 1, 0, 0, 1, 0)
Path B: c(1, 1, 1, 1, 0, 0, 0, 1)
Path C: c(1, 1, 1, 0, 0, 0, 1, 0)
Path D: c(1, 1, 1, 0, 0, 0, 0, 1)

```

These correspond to pmsimstats2025 Paths 1–4 respectively. Tracing the accumulation algorithm for each path yields:

Table 1: Original **tsd** values (in weeks) for all four Hybrid design paths, computed by **buildtrialdesign()**. On-drug rows show **tsd** = 0; off-drug rows show cumulative interval widths. The minimum off-drug **tsd** is 1 (clustered phase) or 4 (spaced phase).

Path	Week							
	4	8	9	10	11	12	16	20
A (ondrug)	1	1	1	1	0	0	1	0
A (tsd)	0	0	0	0	1	2	0	4
B (ondrug)	1	1	1	1	0	0	0	1
B (tsd)	0	0	0	0	1	2	6	0
C (ondrug)	1	1	1	0	0	0	1	0
C (tsd)	0	0	0	1	2	3	0	4
D (ondrug)	1	1	1	0	0	0	0	1
D (tsd)	0	0	0	1	2	3	7	0

3.3 Original Dbc in the Analysis Model

In **lme_analysis.R** (lines 107–109), the original computes **Dbc** as:

```
data.m2[Db==TRUE, Dbc := 1]
data.m2[Db==FALSE, Dbc := (1/2)^(sf * tsd / t_half)]
```

where `sf` is `op$carryover_scalefactor`, which defaults to 1 (not 7 as previously stated in an earlier version of this document). Note that the data-generating process in `generateData()` uses a separate `scalefactor` parameter that defaults to 2 (see Section 6).

3.4 Original Dbc at Boundary Measurements

With the original `tsd` values, the first off-drug measurement in the clustered phase has $t_{sd} = 1$ week:

Table 2: Original Dbc at off-drug measurements for a path C participant, using the original interval-based `tsd`. Boundary measurements carry negligible carryover for short half-lives.

Week	Orig <code>tsd</code>	Orig Dbc ($t_{1/2} = 0.1$)	Orig Dbc ($t_{1/2} = 0.2$)
10	1	$(1/2)^{10} \approx 0.001$	$(1/2)^5 = 0.031$
11	2	$(1/2)^{20} \approx 0$	$(1/2)^{10} \approx 0.001$
12	3	$(1/2)^{30} \approx 0$	$(1/2)^{15} \approx 0$
20	4	$(1/2)^{40} \approx 0$	$(1/2)^{20} \approx 0$

For $t_{1/2} = 0.1$ weeks, the boundary Dbc is 0.001—essentially zero carryover. Even at $t_{1/2} = 0.2$ weeks, the boundary value is only 0.031. These values produce a smooth, gradual decline in the on/off contrast as the half-life increases, not a step function.

4 The 2025 Reimplementation Discrepancy

4.1 The `carryover_power_simulation.R` Algorithm

The 2025 code computes `tsd` as the difference between the current calendar week and the week at which a treatment transition (on $\rightarrow$ off) is detected:

```
just_discontinued = treatment == 0 &
  lag(treatment, default = 0) == 1
discontinuation_week = if_else(
  just_discontinued, week, NA_real_
)
discontinuation_week = zoo::na.locf(
  discontinuation_week, na.rm = FALSE
)
tsd = if_else(
  treatment == 0 & !is.na(discontinuation_week),
  pmax(week - discontinuation_week, local_tsd_min),
  0
)
```

The implicit assumption is that discontinuation occurs *at* the measurement week. At the first off-drug measurement, `week == discontinuation_week`, so $t_{sd} = 0$.

4.2 Side-by-Side $\mathbf{tsd}$ Comparison

Table 3 compares the original and 2025 $\mathbf{tsd}$ values for all four paths. Only off-drug rows are shown.

Table 3: Original vs. 2025 $\mathbf{tsd}$ at off-drug measurements. The 2025 value is systematically one interval width shorter at the first off-drug measurement after each discontinuation. Boldface marks mismatches.

Path	Week	Orig $\mathbf{tsd}$	2025 $\mathbf{tsd}$	Match?
A	11	1	0	No
A	12	2	1	No
A	20	4	0	No
B	11	1	0	No
B	12	2	1	No
B	16	6	5	No
C	10	1	0	No
C	11	2	1	No
C	12	3	2	No
C	20	4	0	No
D	10	1	0	No
D	11	2	1	No
D	12	3	2	No
D	16	7	6	No

Every off-drug measurement is affected. The offset equals the interval width leading into the first off-drug timepoint: 1 week in the clustered phase (weeks 10–12) and 4 weeks in the spaced phase (weeks 16, 20). Re-discontinuation points (e.g., week 20 for paths A and C) show the largest discrepancy: original $\mathbf{tsd} = 4$ versus 2025 $\mathbf{tsd} = 0$.

4.3 Consequence for Dbc

Table 4 shows the resulting Dbc values for a path C participant at $t_{1/2} = 0.1$ weeks.

Table 4: Dbc under original vs. 2025 $\mathbf{tsd}$ for path C, $t_{1/2} = 0.1$ weeks. The 2025 code produces full carryover ($\mathbf{Dbc} = 1$) at boundary measurements where the original produces negligible carryover.

Week	Orig $\mathbf{tsd}$	Orig Dbc	2025 $\mathbf{tsd}$	2025 Dbc	Discrepancy
10	1	0.001	0	1.000	$\times 1000$
11	2	≈ 0	1	0.001	—
12	3	≈ 0	2	≈ 0	—
20	4	≈ 0	0	1.000	∞

At $t_{1/2} = 0.2$ weeks the pattern is the same:

Table 5: Dbc under original vs. 2025 `tsd` for path C, $t_{1/2} = 0.2$ weeks.

Week	Orig <code>tsd</code>	Orig Dbc	2025 <code>tsd</code>	2025 Dbc	Discrepancy
10	1	0.031	0	1.000	$\times 32$
11	2	0.001	1	0.031	$\times 31$
12	3	≈ 0	2	0.001	—
20	4	≈ 0	0	1.000	∞

5 The Step-Function Power Cliff (a 2025 Bug)

5.1 Mathematical Origin

The step-function power cliff observed in the 2025 simulation arises because $t_{sd} = 0$ at boundary measurements. The exponential decay model evaluates as:

$$f(t_{1/2})|_{t_{sd}=0} = \begin{cases} 0 & \text{if } t_{1/2} = 0 \quad (\text{special case in code}) \\ \left(\frac{1}{2}\right)^{0/t_{1/2}} = \left(\frac{1}{2}\right)^0 = 1 & \text{if } t_{1/2} > 0 \end{cases}$$

This is a step function: for **any** positive half-life, no matter how small, the carryover effect at the boundary equals 1. The parameter $t_{1/2}$ controls only the rate of subsequent decay, not the initial value. This behavior is specific to the 2025 `tsd` computation and does not occur in the original, where $t_{sd} \geq 1$ at all boundary measurements.

5.2 Consequences for the 2025 Simulation

In the 2025 simulation with `tsd_min` = 0 (uncorrected), the naive model (`treatment`) shows a step-function collapse:

Table 6: 2025 simulation results with uncorrected `tsd` (`tsd_min` = 0, $N = 70$, 200 iterations). The naive model collapses to ~ 0.06 power at any nonzero half-life. This behavior is caused by the `tsd` bug, not by a property of the carryover decay model.

$t_{1/2}$ (wks)	Power		Mean $\hat{\beta}$		Difference (power)
	Dbc	Treatment	Dbc	Treatment	
0.00	0.73	—	0.675	—	—
0.05	0.63	0.07	0.647	0.144	+0.56
0.10	0.69	0.06	0.694	0.133	+0.64
0.15	0.66	0.06	0.667	0.139	+0.60
0.20	0.57	0.07	0.631	0.124	+0.50

The mechanism: at any nonzero $t_{1/2}$, weeks 10 and 20 (and their equivalents across all four paths) carry `Dbc` = 1—full drug effect—because the 2025 code assigns $t_{sd} = 0$. These boundary measurements constitute half of all off-drug timepoints. The naive model assigns `treatment` = 0 to these timepoints while they still carry full biomarker-moderated drug response, collapsing the on/off contrast. The mean interaction estimate drops from $\hat{\beta} \approx 0.68$ to $\hat{\beta} \approx 0.13$ —an 80% attenuation.

5.3 The `tsd_min` Workaround

As part of the 2025 development, a minimum `tsd` floor was introduced to address the step-function cliff:

$$t_{\text{sd,eff}} = \max(t_{\text{sd}}, t_{\text{sd,min}})$$

With $t_{\text{sd,min}} = 0.5$ weeks, the boundary carryover becomes a smooth function of $t_{1/2}$:

Table 7: Boundary carryover under three regimes: the buggy 2025 code (`tsd` = 0), the `tsd_min` = 0.5 workaround, and the original interval-based `tsd` = 1 (clustered phase).

$t_{1/2}$ (wks)	2025 bug ($t_{\text{sd}} = 0$)	Workaround ($t_{\text{sd}} = 0.5$)	Original ($t_{\text{sd}} = 1$)
0.00	0	0	0
0.05	1.000	0.001	≈ 0
0.10	1.000	0.031	0.001
0.15	1.000	0.101	0.010
0.20	1.000	0.177	0.031

The workaround eliminates the step function and produces gradual power decline:

Table 8: 2025 simulation results with `tsd_min` = 0.5 ($N = 70, 200$ iterations). The step function is eliminated, but these results do not match the original’s `tsd` computation, where the effective minimum is 1 week (clustered) or 4 weeks (spaced).

$t_{1/2}$ (wks)	Power		Mean $\hat{\beta}$		Difference (power)
	Dbc	Treatment	Dbc	Treatment	
0.00	0.69	—	0.646	—	—
0.05	0.68	0.72	0.663	0.669	−0.04
0.10	0.68	0.73	0.653	0.656	−0.05
0.15	0.66	0.58	0.653	0.579	+0.08
0.20	0.53	0.54	0.625	0.544	−0.01

While the workaround eliminates the step-function artifact, it does not reproduce the original’s `tsd` values. Table 7 shows that the workaround ($t_{\text{sd}} = 0.5$) produces systematically higher boundary carryover than the original ($t_{\text{sd}} = 1$) at every half-life. The correct fix is to adopt the original’s interval-based accumulation logic (Section 7).

6 Open Question: The Original’s Power Behavior

Independent of the 2025 `tsd` bug, the published Hendrickson heatmaps [1] show power declining with increasing carryover half-life. Whether this decline is steeper than pharmacologically expected is the motivating question for the `carryover_focus/` analysis. Once the `tsd` bug is fixed, any remaining excessive power dropoff must have a different mechanism. Three candidates merit investigation.

6.1 Scale Factor Mismatch Between DGP and Analysis

The original code uses different scale factors for carryover in the data-generating process and the analysis model:

- **DGP** (`generateData.R`, line 96): `scalefactor = 2` (default).

$$\text{carryover}_{\text{DGP}} = \text{br}_{\text{prev}} \times \left(\frac{1}{2}\right)^{2 \cdot t_{\text{sd}}/t_{1/2}}$$

- **Analysis** (`lme_analysis.R`, line 109): `carryover_scalefactor = 1` (default).

$$\text{Dbc} = \left(\frac{1}{2}\right)^{1 \cdot t_{\text{sd}}/t_{1/2}}$$

The DGP generates faster decay (higher effective exponent) than the analysis model assumes. At $t_{\text{sd}} = 1$, $t_{1/2} = 0.2$:

- DGP carryover: $(1/2)^{2 \times 1/0.2} = (1/2)^{10} \approx 0.001$
- Analysis Dbc: $(1/2)^{1 \times 1/0.2} = (1/2)^5 = 0.031$

The analysis model overestimates the residual drug effect by a factor of ~ 30 at this timepoint. This systematic misspecification could contribute to power loss by distorting the interaction contrast that Model A attempts to estimate.

6.2 Cumulative Carryover in `generateData()`

In the DGP (`generateData.R`, lines 92–99), carryover is added to `brmeans` cumulatively:

```
for (p in 2:nP) {
  if (!d[p]$onDrug) {
    if (d[p]$tsd > 0) {
      brmeans[p] <- brmeans[p] +
        brmeans[p-1] * (1/2)^(sf * d$tsd[p] / t_half)
    }
  }
}
```

This recursive structure means the biological response at off-drug timepoints depends on the entire preceding on-drug trajectory, not just the value at discontinuation. The interaction with the Gompertz response curve (`modgompertz()`) and the multi-factor correlation structure may produce nonlinear contamination patterns that the analysis model cannot fully capture.

6.3 Correlation Structure Interaction

The original DGP generates correlated random components across timepoints via a multivariate normal draw with a structured correlation matrix (autocorrelation within factors, cross-correlation between factors). The carryover adjustment modifies the mean structure but not the correlation matrix. As $t_{1/2}$ increases, the mismatch between the adjusted means and the original correlation structure grows, potentially producing model misspecification that reduces the interaction estimate.

6.4 Status

These three candidates are not mutually exclusive. Determining their relative contributions requires running the corrected 2025 simulation (with original-matching `tsd`) and comparing its power curves

against the published results. The scale factor mismatch (Section 6.1) is the most straightforward to test: it can be isolated by running simulations with matched scale factors (both set to 1 or both set to 2) and observing whether the power curve changes shape.

7 Recommended Path Forward

1. **Fix `tsd` in `carryover_power_simulation.R`.** Replace the calendar-subtraction logic with the original’s interval-based accumulation. Verify that the corrected code produces identical `tsd` values to `buildtrialdesign()` for all four paths (Table 1).
2. **Re-run the two-scenario comparison** with the corrected `tsd` and `tsd_min = 0` to establish a faithful reproduction of Hendrickson’s simulation as a baseline.
3. **Compare power curves** from the corrected simulation against the published heatmaps. If power still drops more rapidly than expected, investigate the scale factor mismatch by running with matched scale factors.
4. **Evaluate `tsd_min` as a design modification.** If the original’s power behavior proves excessive, a minimum washout interval represents a pharmacologically defensible improvement: in clinical practice, there is always a nonzero gap between the last dose and the next assessment. This should be evaluated as a deliberate modeling choice with its own simulation results, not as a bug fix.

8 Conclusions

1. **The step-function power cliff in the 2025 simulation is caused by a `tsd` computation bug.** The 2025 code computes $t_{sd} = \text{week} - \text{discontinuation_week}$, producing $t_{sd} = 0$ at boundary measurements. The original computes `tsd` as cumulative off-drug interval widths, producing $t_{sd} \geq 1$ at all off-drug timepoints. The step function does not exist in the original framework.
2. **The `tsd_min` workaround partially compensates but does not match the original.** With $t_{sd,min} = 0.5$, the boundary carryover at $t_{1/2} = 0.1$ is 0.031; with the original `tsd`, it is 0.001. The correct fix is to adopt the original’s interval-based accumulation.
3. **The original’s power behavior is a separate, open question.** The published heatmaps show power declining with increasing $t_{1/2}$. Whether this decline is excessive requires investigation after the `tsd` bug is fixed. Leading candidates include the scale factor mismatch between the DGP (`scalefactor = 2`) and the analysis model (`carryover_scalefactor = 1`), and the cumulative carryover structure in `generateData()`.
4. **The two-scenario comparison framework remains valid.** Comparing Hendrickson’s Dbc predictor against the naive `treatment` indicator is the right experimental design. The simulation results in Tables 6–8 document the behavior of the buggy code and its workaround; new results with the corrected `tsd` are needed.

References

- [1] Hendrickson RC, Thomas RG, Schork NJ. Optimizing aggregated N-of-1 trial designs for predictive biomarker validation: Statistical methods and theoretical findings. *Frontiers in Digital Health*. 2020;2:13. doi: [10.3389/fdgth.2020.00013](https://doi.org/10.3389/fdgth.2020.00013)
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- [3] Yap C, Wang X, Harhay MO. Behavioral carry-over effect and power consideration in crossover trials. *Biometrics*. 2024;80(2):ujae023. doi: [10.1093/biometcs/ujae023](https://doi.org/10.1093/biometcs/ujae023)
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- [5] Senn S. Sample size considerations for n-of-1 trials. *Statistical Methods in Medical Research*. 2019;28(2):372–383. doi: [10.1177/0962280217726801](https://doi.org/10.1177/0962280217726801)

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